

From Functions to Agents: Architecting a Delay-Aware Agentic Core for 6G Mobile Networks

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Abstract—To cope with increasing variability in service demands and operational conditions beyond 5G, the mobile core control plane must support faster and more context-aware decision making under strict delay constraints. This paper examines the architectural feasibility of introducing bounded agent-based decision components into the mobile core, with control-loop delay treated as the primary design constraint across all control timescales. It proposes a delay-structured Agentic Core architecture as an evolution path of the Service-Based Architecture (SBA), in which agentic behavior is incrementally integrated while preserving SBA execution semantics in early stages. Four implementation stages are outlined, ranging from analytics-triggered playbooks to agent-native control, and are compared in terms of decision latency, integration depth, and architectural impact. By explicitly structuring control and coordination along delay domains, the paper highlights how delay-aware integration is essential for extending adaptability in future 6G mobile core networks.

Index Terms—Delay-Aware Control, Agentic Core Architecture, 6G, 6G Core Network, SBA Evolution

I. INTRODUCTION

Mobile core networks were historically designed under assumptions of predictable traffic patterns and bounded failure modes, resulting in architectures composed of standardized Network Functions (NFs) that execute finite-state logic and derive control decisions from administratively pre-configured policies [1]. While this approach has provided robustness and operational stability, it is becoming increasingly misaligned with networks beyond 5G, where service demands, deployment environments, and traffic dynamics exhibit higher variability, tighter latency budgets, and growing pressure for cost-efficient operations. This increasingly stresses the core connectivity control itself, which must react within tighter and more localized timing bounds [2].

Within the mobile core, control decisions constitute the most delay-sensitive and structurally fragile elements of the system. Operators must continuously coordinate radio access, transport, and compute resources under volatile conditions while maintaining millisecond-scale control-plane response times. In practice, this coordination is driven primarily by administratively defined policies that encode expected operating conditions and response rules to address normal and exceptional cases alike. While effective under nominal assumptions, policy-driven control remains constrained by predefined decision boundaries and typically adapts at coarse timescales. As network dynamics intensify, rapid context-aware reaction

is increasingly replaced by conservative policy margins and overprovisioning, leading to rising operational costs.

The introduction of the Service-Based Architecture (SBA) in 5G improved modularity and interface-level flexibility by decomposing the core into loosely coupled microservice-based functions. However, while SBA enables elastic instantiation and scalable service exposure, it preserves static decision logic within each NF. In congestion or exceptional conditions, operators would replicate entire NF instances, increasing compute and memory consumption without reducing decision latency or modifying control behavior [1]. As a result, infrastructure-level elasticity does not translate into delay-aware adaptability at the control-plane level.

This structural limitation leads to a recurring paradox: control policies tuned for expected operating conditions degrade rapidly when assumptions are shattered. Under flash crowds, cascading contention, or unanticipated failures, service degradation is often driven not by resource exhaustion, but by delayed or inflexible control responses [3], [4]. In such situations, bounded reaction delay becomes a primary determinant of system resilience.

To address this challenge, this article investigates an architectural evolution in which bounded agent-based decision components are progressively introduced into the mobile core control plane, with control-loop delay treated as the primary design constraint. The proposed Agentic Core architecture is explicitly structured along delay domains, separating reflexive control, coordination, and planning according to their respective timing budgets, and enabling incremental integration without immediate disruption of existing deployments.

By structuring control logic, coordination, and adaptation strictly according to timing domains, the work provides a systematic way to reason about where agentic behavior is feasible in mobile core networks and where it is not. This delay-centric perspective enables comparative analysis across architectural options, clarifies trade-offs between adaptability and control-loop latency, and establishes a concrete basis for assessing agent integration under carrier-grade constraints.

The remainder of the article is structured as follows. Section II reviews related work while Section III presents the delay-structured Agentic Core architecture, followed by its implementation options in Section IV. Protocol considerations are discussed in Section V. Section VI provides a feasibility assessment, and conclusions in Section VII.

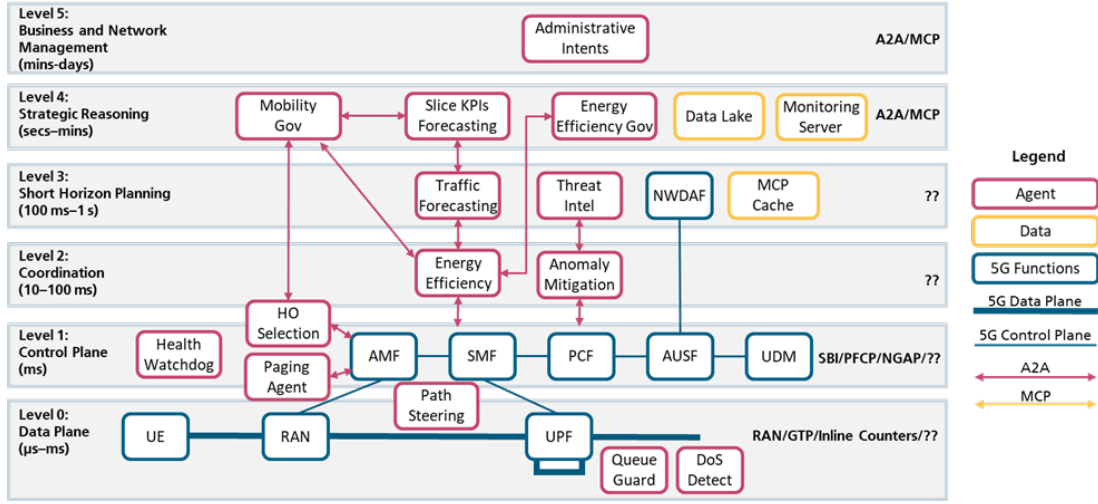


Fig. 1. Layered control structure in the Agentic Core with example agents and communication interfaces.

II. BACKGROUND

The 5G mobile core is standardized around the SBA, where NFs implement control logic with well-defined roles, interfaces, and procedures [1]. These functions execute deterministic control flows derived from finite-state machines and policy evaluation rules, ensuring predictable behavior, interoperability, and compliance with carrier-grade timing constraints. While this design is essential for stable operation, it limits the degree to which control behavior can be adapted dynamically.

Although the service-based decomposition enables flexible deployment and horizontal scaling, it does not fundamentally change how control decisions are executed. Adaptation in today's 5G core is achieved through configuration updates, policy tuning, or instance replication, rather than through runtime modification of decision logic. As a result, scalability at the infrastructure level does not directly translate into faster or more context-aware control under dynamic conditions.

To enhance observability and indirect adaptation, 3GPP introduced complementary mechanisms such as the Network Data Analytics Function (NWDAF), which provides analytics and event notifications to control functions [5], and intent-based frameworks that translate operator objectives into configuration actions [6]. These mechanisms improve situational awareness but remain external to the execution path of control procedures and operate at slower timescales, leaving the fundamental timing characteristics of control decisions unchanged.

In parallel, driven by the proliferation of Large Language Models (LLMs), agent-based control approaches have been studied outside the mobile networking domain as a means to enable distributed coordination and adaptive decision-making. Such systems commonly rely on explicit communication substrates, including Agent-to-Agent (A2A) protocols for direct interaction between decision entities and Model Context Protocols (MCP) for structured access to shared models and contextual information [7]. While these approaches offer flexibility and expressiveness, they are generally designed for

asynchronous execution and very soft latency constraints, which are incompatible with the close to deterministic timing requirements of the 5G core.

Nevertheless, the ability of agent-based systems to structure decision-making across heterogeneous timescales aligns with emerging control challenges in mobile networks [8], [9]. The open issue is whether such mechanisms can be sufficiently constrained to operate within the delay, interoperability, and deployment boundaries defined by current 5G standards. This question motivates the architectural analysis and feasibility assessment presented in the following sections.

III. SYSTEM ARCHITECTURE

To support agentic behavior under carrier-grade constraints, the mobile core architecture must be structured primarily around delay, rather than around functional decomposition alone. In the absence of such structuring, interactions between fast-reacting control logic and slower reasoning components risk introducing feedback delays, unstable coordination, or violations of real-time execution requirements. The Agentic Core therefore, organizes control functionality explicitly along timing domains, ensuring that each decision component operates only within the delay envelope it can reliably satisfy.

As illustrated in Figure 1, the architecture is composed of a set of logical layers, each associated with a characteristic reaction time, scope of observability, and control responsibility. The defining architectural principle is latency containment: control decisions are allowed to influence only layers operating at equal or slower timescales, while upstream propagation toward faster layers is explicitly avoided. This prevents long-horizon reasoning from interfering with time-critical control paths, while still enabling gradual adaptation through parameterization and intent propagation.

At the lowest layer, agentic logic is co-located with data-plane elements such as the UPF. Decisions at this level operate within microsecond to sub-millisecond bounds and act directly on local state, including queues, counters, and flow metadata.

These reflexive agents implement immediate mitigation actions, such as rate limiting or anomaly suppression, without invoking control-plane signaling. Their behavior remains strictly localized, ensuring that data-plane responsiveness is preserved even under overload or partial failure conditions.

The next layer introduces agentic components embedded in or adjacent to control-plane NFs, including Access and Mobility Function (AMF) and Session Management Function (SMF). Decisions at this level must be completed within a few milliseconds and are therefore constrained to a limited context and bounded reasoning complexity. Typical actions include mobility-related choices, session steering, or localized health reactions. Execution remains compatible with SBA semantics, either by operating within NF logic or by invoking fast, synchronous service interactions, thereby preserving deterministic control behavior.

Coordination across multiple NFs and domains is handled at the subsequent layer, where control latency budgets extend into the tens of milliseconds. Agents at this level aggregate observations from lower layers and perform distributed arbitration, such as slice admission decisions, anomaly correlation, or energy-aware scaling. Interaction is predominantly horizontal and peer-based, relying on low-delay A2A communication substrates to exchange soft state and coordination signals. While decisions are no longer strictly local, they remain bounded in scope to prevent cascading delay amplification.

Beyond immediate coordination, short-horizon planning functions operate at sub-second timescales. Agents at this layer analyze recent telemetry, perform forecasting, and adjust control parameters that influence lower layers indirectly. MCPs are used to share model state, thresholds, and contextual information, enabling consistency across planning agents without imposing synchronous dependencies on faster control paths. Actions are expressed as parameter updates rather than imperative commands, preserving stability under fluctuating conditions.

At the highest operational layer, strategic reasoning spans seconds to minutes and focuses on longer-term objectives such as KPI alignment, energy budgets, or compliance constraints. Agents at this level observe aggregated system behavior over extended windows and synthesize policy guidance that shapes the operating envelopes of lower layers. Their influence is intentionally indirect, ensuring that long-horizon optimization does not compromise short-term responsiveness.

An additional administrative layer contextualizes the system with business objectives and operational policies but remains outside the real-time control loop. While it may issue high-level intents or evaluate system conformance, it does not participate in delay-critical decision paths.

Table I complements the architectural view by listing representative agents per timing tier, together with their latency targets and operational roles. The examples are illustrative rather than exhaustive, but they demonstrate how agentic behavior can be composed incrementally across timescales while respecting strict delay boundaries. Together, the layered structure and timing discipline provide a concrete architectural

basis for assessing where agent-based control is feasible in mobile core networks and where conventional deterministic logic must remain dominant.

IV. AGENTIC CORE IMPLEMENTATION STAGES

To translate the delay-structured architecture into deployable systems, the Agentic Core is realized through a set of incremental implementation stages that integrate agentic logic progressively deeper into existing control execution paths. These stages are not competing designs, but successive points along a migration continuum, each characterized by a specific balance between achievable decision latency, architectural disruption, and operational risk. The ordering of the stages reflects an explicit feasibility logic: approaches that preserve existing execution semantics and timing guarantees are introduced first, while approaches that relax these constraints are considered only where delay budgets permit.

The first stage introduces agentic logic exclusively outside the control execution path. Analytics outputs are enriched and interpreted by agent-assisted logic that refines or selects among predefined policy actions, while control procedures themselves remain fully deterministic and unchanged. Decision execution continues to follow established control flows, and no additional delay is introduced into time-critical paths. As a result, overall control-loop latency continues to be dominated by existing signaling procedures, typically on the order of hundreds of milliseconds. This option is fully compatible with current deployments and primarily serves as a low-risk entry point for agent-assisted reasoning without affecting carrier-grade timing guarantees.

The second stage embeds agentic logic directly within the execution scope of individual control functions. Here, agent components operate on local context and participate in decision making within millisecond-scale timing budgets. Execution semantics remain aligned with established service interactions, but internal decision rules become adaptive based on recent observations and short-term context. Because reasoning remains localized and bounded, delay amplification is limited, while decision latency can be reduced compared to externally driven adaptation. This option preserves interoperability and control determinism while enabling delay-aware adaptation at the granularity of individual functions.

In the third stage, agentic components assume the internal functional role of control functions themselves. External interfaces continue to expose standard service endpoints, ensuring compatibility at system boundaries, but internal execution is no longer organized around monolithic function logic. Instead, decision-making is distributed across coordinated agent collectives that interact horizontally through low-latency communication. This enables faster cross-functional coordination and more flexible behavior under dynamic conditions, but requires explicit control of coordination delay to avoid instability. Here, compatibility is preserved at the interface level, while internal execution diverges from conventional control realizations.

The final stage represents a full transition to agent-native control execution. All control logic is realized through spe-

TABLE I
EXAMPLES OF POTENTIAL AGENTS IN THE 5G/6G CORE NETWORK

Timing Tier (latency target)	Representative Agent	FCAPS Focus	One-line Job Description
L0 Data-plane reflex (μ s – sub-ms)	Queue-Guard agent CDR Rate-Limiter Inline DoS Sentinel	P A S	Marks packets ECN / RED when UPF queue watermark trips. Throttles duplicate/burst CDRs before storage. Drops or gray-lists SUPIs that exceed attach-rate thresholds.
L1 Control-plane reflex (1 – 10 ms)	Mobility-Selection μ -agent Path-Steering μ -agent Fast-Health Watchdog	P P F	Picks best paging-area / target cell on the fly. Chooses the lowest-cost UPF and tunnel for a new PDU session. Fires fast alerts on CPU/heap spikes inside an NF pod.
L2 Coordination mesh (10 – 100 ms, A2A bus)	Slice-Broker Anomaly-Mitigator Energy-Efficiency Advisor	C / P F / P P	Admits / rejects slice requests; pushes guard-bands to μ -agents. Correlates fast alerts to auto-scale or restart NFs via k8s. Identifies idle UPFs, issues shutdown-candidate tasks.
L3 Short-horizon planning (0.1 – 1 s, MCP bus)	Threat-Intel Insight Traffic-Forecasting agent Billing-Anomaly Detector	S P A	Pulls CVE / IOC feeds; distributes block rules to Sec-agents. Predicts next-day load, feeds Slice-Broker and Capacity-Planner. Flags fraud or tap-vs-CDR gaps in near-real-time stream.
L4 Strategic reasoning (seconds – minutes)	Multi-Slice KPI Forecaster Energy-Efficiency Governor Compliance Auditor	P P S / G	Runs RL or simulation to predict 1 h SLA risk; sets guard-bands. Keeps monthly energy / CO2 budget; guides Advisor and Planner. Periodically audits LI hooks and config drift; issues signed reports.

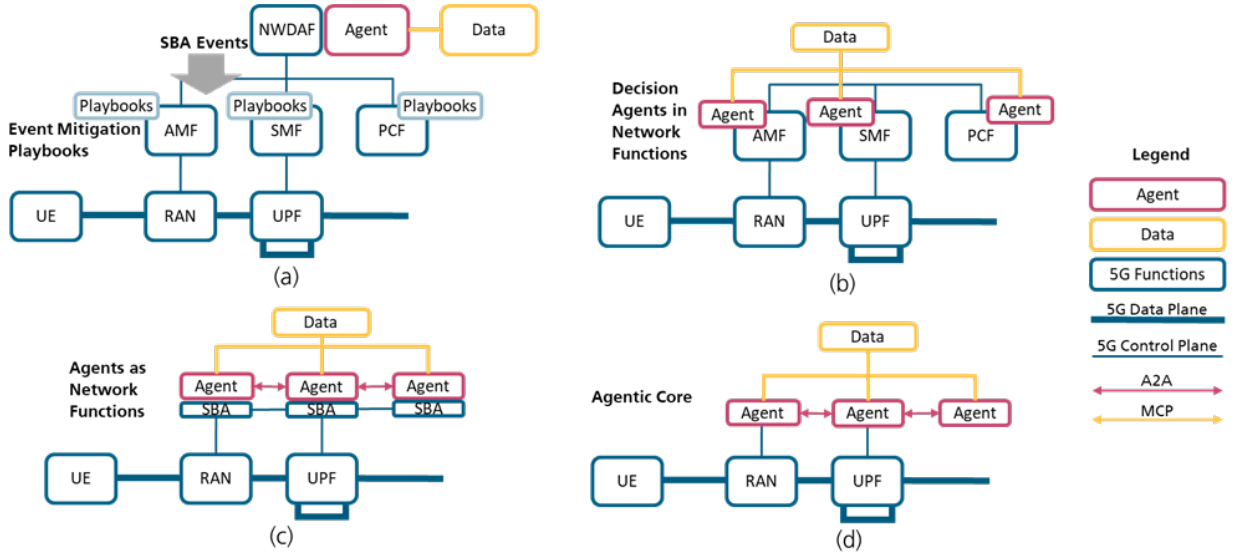


Fig. 2. Incremental implementation stages for Agentic Core deployment.

cialized agents operating across the defined timing layers, and control loops are formed dynamically based on observed conditions and objectives. Coordination spans reflexive response, short-range arbitration, and longer-horizon planning, enabling unified adaptation across timescales. However, this option abandons internal service-based execution semantics, retaining compatibility only at interworking boundaries. As such, it is viable only where strict delay containment can be guaranteed and architectural disruption is acceptable.

Across all stages, feasibility is determined primarily by whether agentic reasoning can be constrained to operate within the delay envelope of the control tier in which it is deployed. Early options prioritize preservation of existing timing guarantees, while later options trade architectural continuity for faster coordination and broader adaptability. This staged approach provides a concrete basis for assessing where agentic control

can be introduced incrementally in mobile core networks, and where conventional deterministic execution must remain dominant.

V. PROTOCOL CONSIDERATIONS

The introduction of agentic control into the mobile core requires a careful alignment between control semantics and protocol behavior, as different interaction models impose fundamentally different delay, coupling, and execution characteristics. SBA, A2A, and MCP each embody distinct assumptions regarding latency tolerance, state handling, and lifecycle integration. Selecting an appropriate protocol per control tier is therefore essential to ensure that agentic behavior remains compatible with carrier-grade timing constraints.

Service-based interactions provide synchronous, stateless execution semantics with well-defined procedure boundaries.

Built on request–response signaling and standardized service exposure, they support millisecond-scale latency and deterministic control behavior. These properties make service-based signaling suitable for delay-critical decision paths, where agentic logic is embedded directly into existing execution flows and must preserve established timing guarantees. At these layers, agentic behavior remains constrained by the same procedural limits as conventional control logic.

Peer-oriented interaction mechanisms enable more flexible coordination across distributed decision entities. A2A communication supports horizontal information exchange and decentralized arbitration without centralized orchestration [10]. MCP complement this by providing structured access to shared model state and contextual information during execution [11]. Both mechanisms favor soft-state interaction and tolerate transient or eventually consistent views, which introduces additional delay and variability. As a result, their use is feasible only at control layers where delay budgets extend beyond strict millisecond bounds and where indirect influence on control behavior is acceptable.

To support coordination within tighter timing envelopes, adaptations are required. Persistent transport sessions, compact message encoding, and agent co-location can reduce communication overhead. MCP interactions may further benefit from schema caching, replica-local model access, and reduced negotiation depth [12]. These optimizations improve responsiveness but trade off traceability and introspection, requiring architectural containment to prevent unintended interference with faster control paths.

Across the implementation stages described earlier, protocol usage evolves accordingly. Early stages rely almost exclusively on service-based signaling to preserve interoperability and timing guarantees. As coordination spans multiple functions and domains, peer-oriented interactions are introduced selectively at layers where their latency characteristics remain acceptable. Fully agent-native execution expands the role of peer and model-oriented interaction but only under conditions where strict delay containment is enforced, and internal execution semantics are allowed to diverge from conventional service-based models.

In practice, hybrid protocol usage is unavoidable. Delay-critical decision paths continue to rely on tightly bounded service interactions, while agent coordination and planning leverage more flexible communication substrates at higher timing layers. This alignment between protocol choice and delay domain establishes the foundation for assessing feasibility, as it defines the limits where agentic control can be introduced without violating carrier-grade control-plane requirements.

VI. FEASIBILITY ASSESSMENT

The feasibility of introducing agentic control into the mobile core cannot be evaluated in isolation from broader questions of behavioral correctness, learning stability, and operational governance. These aspects constitute fundamental challenges for adaptive control systems and remain active research topics beyond the scope of this paper. The feasibility assessment

presented here is therefore explicitly conditional: it evaluates whether, assuming agentic behavior can be sufficiently bounded and operationally admissible, such behavior can be integrated into the mobile core without violating carrier-grade delay constraints, deployment models, and interoperability assumptions.

Under this premise, control-loop delay emerges as the dominant feasibility gate. Independent of how agentic behavior is realized internally, any decision component introduced into the mobile core must operate within the strict timing envelopes imposed by access procedures, session control, and data-plane interaction. As shown by the proposed delay-structured architecture, agentic logic is feasible only when its execution time, communication dependencies, and scope of influence are compatible with the delay budget of the control tier in which it is deployed. This constraint applies uniformly across all implementation options and immediately excludes remote coordination, asynchronous inference, or unbounded reasoning from delay-critical control paths.

At data-plane and fast control-plane layers, where decisions must complete within microsecond to low-millisecond bounds, feasibility is limited to strictly local and computationally bounded agentic logic. In these tiers, agentic behavior can refine static thresholds or selection heuristics but cannot introduce additional signaling, coordination latency, or structural changes to control flows. At higher tiers, where delay budgets extend into tens or hundreds of milliseconds, feasibility expands to include agent-to-agent coordination and indirect influence on control behavior, provided that such interaction remains explicitly contained and does not propagate back into time-critical execution paths. This tier-dependent feasibility holds regardless of the sophistication of the agent logic and is dictated solely by timing constraints.

Beyond timing, architectural and deployment feasibility further constrain practical adoption. Implementation options that preserve existing Network Function boundaries and service-based execution semantics align with current cloud-native core deployments and can be realized using existing containerization, orchestration, and lifecycle management mechanisms. These options support incremental introduction and coexistence with legacy logic. As agentic control assumes broader functional responsibility and spans multiple control functions, feasibility increasingly depends on explicit containment of coordination scope, failure domains, and delay propagation. Fully agent-native control execution is therefore feasible only in environments where internal execution semantics may diverge from function-centric realizations and where architectural redesign is acceptable.

An additional feasibility condition concerns the admissibility of agentic decisions before they are applied to the live control plane. Agent-generated actions cannot be assumed to be correct or safe by construction, particularly as decision scope increases at higher control tiers. Feasible deployment therefore requires that agent outputs are subject to explicit guardrails whose strength scales with available delay budget. At fast control layers, such guardrails must be lightweight and

local, enforcing syntactic, semantic, or range-based checks that preserve timing guarantees. At slower coordination and planning layers, where additional delay can be tolerated, agent behavior may be evaluated against digital twins, simulations, or predictive system models before changes are propagated. This progressive validation approach aligns naturally with the delay-structured architecture: the more time available for decision making, the stronger the admissible verification mechanisms, without allowing unverified behavior to influence delay-critical control paths.

Importantly, this assessment does not claim that satisfying delay feasibility and guardrail enforcement is sufficient for safe or optimal operation. Rather, it establishes necessary conditions. If agentic behavior cannot be constrained to operate within the delay envelope of a given control tier, or if its outputs cannot be checked with tier-appropriate guardrails, then integration is infeasible regardless of advances in learning methods or governance frameworks. Conversely, satisfying these conditions defines a feasible architectural space in which the remaining challenges may be addressed incrementally.

Taken together, the feasibility analysis indicates that agentic control in the mobile core is viable only when adaptation is explicitly bounded along delay domains, integrated incrementally into existing execution models, and coupled with tier-appropriate validation mechanisms. By structuring the Agentic Core around these constraints, the paper delineates where agentic behavior can be introduced without violating carrier-grade requirements and where deterministic control must remain dominant. This feasibility envelope provides a concrete basis for architectural decision-making and avoids treating agentic control as an unconstrained or monolithic capability.

VII. CONCLUSIONS AND FURTHER WORK

This paper has examined the architectural feasibility of introducing bounded agent-based decision components into the mobile core control plane under carrier-grade delay constraints. By treating control-loop delay as the primary design constraint, the proposed delay-structured Agentic Core organizes control and coordination along explicit timing domains, thereby containing latency propagation and separating time-critical execution from slower coordination and planning. The resulting architecture provides a concrete basis for reasoning about where agentic behavior can be integrated without violating millisecond-scale control requirements, and where deterministic service-based execution must remain dominant.

The analysis has shown that feasibility is inherently tier-dependent and incremental. At delay-critical layers, only strictly local and computationally bounded agentic logic is admissible, while coordination-oriented tiers can accommodate richer agent interaction provided that communication and influence remain explicitly contained. In parallel, the feasibility assessment has underlined that any practical adoption requires tier-appropriate guardrails, ranging from lightweight local checks in fast control paths to stronger validation mecha-

nisms, including digital twins and simulation-based evaluation, at higher layers where delay budgets permit.

As further work, the next step is to validate the proposed evolution path through a stepwise implementation in the Open6GCore. In this validation, representative control decisions will be progressively migrated through the four stages, starting with agent-assisted policy refinement outside the execution path, continuing with embedded agent components within selected Network Functions, and subsequently evolving toward agent-native realizations while maintaining external interoperability. This step-by-step progression enables controlled integration and supports comparative assessment of agent classes and placement strategies under realistic control-plane timing constraints. The resulting implementation-driven insights are expected to inform practical engineering choices and to provide input for future standardization discussions on delay-aware integration of agentic control in 6G mobile core networks.

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